Review of Block ciphers:
Authenticated Encryption

Study reference: Aumasson, chapter 8

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Recap: encryption does not guarantee integrity!

- **Confidentiality: semantic security against CPA**
  - Encryption secure against eavesdropping only
  - NOT secure against active attackers (see next 5 slides!!)

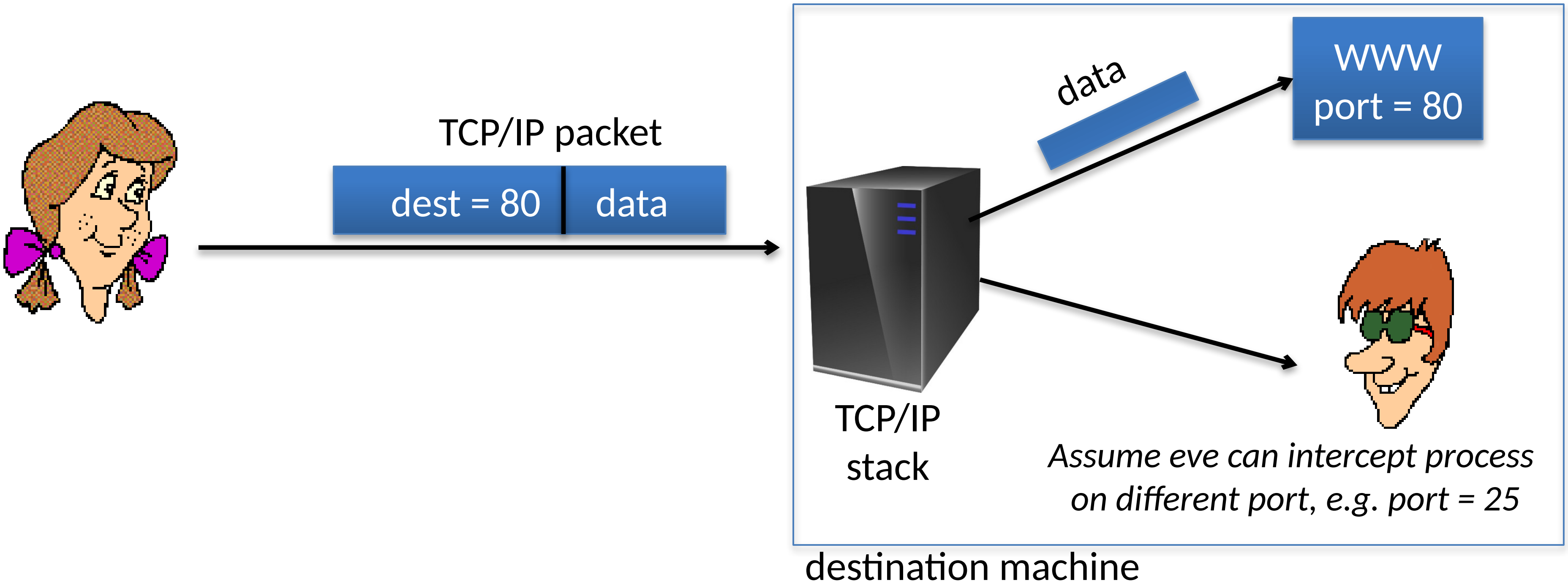
- **Integrity: unforgeability under CCA**
  - But messages remain in plaintext

- **Authenticated Encryption:**
  - Cipher designed to be secure against active attacks (tampering)
  - Ensure both confidentiality and integrity

Indeed: tampering attacks CAN *also* break confidentiality!

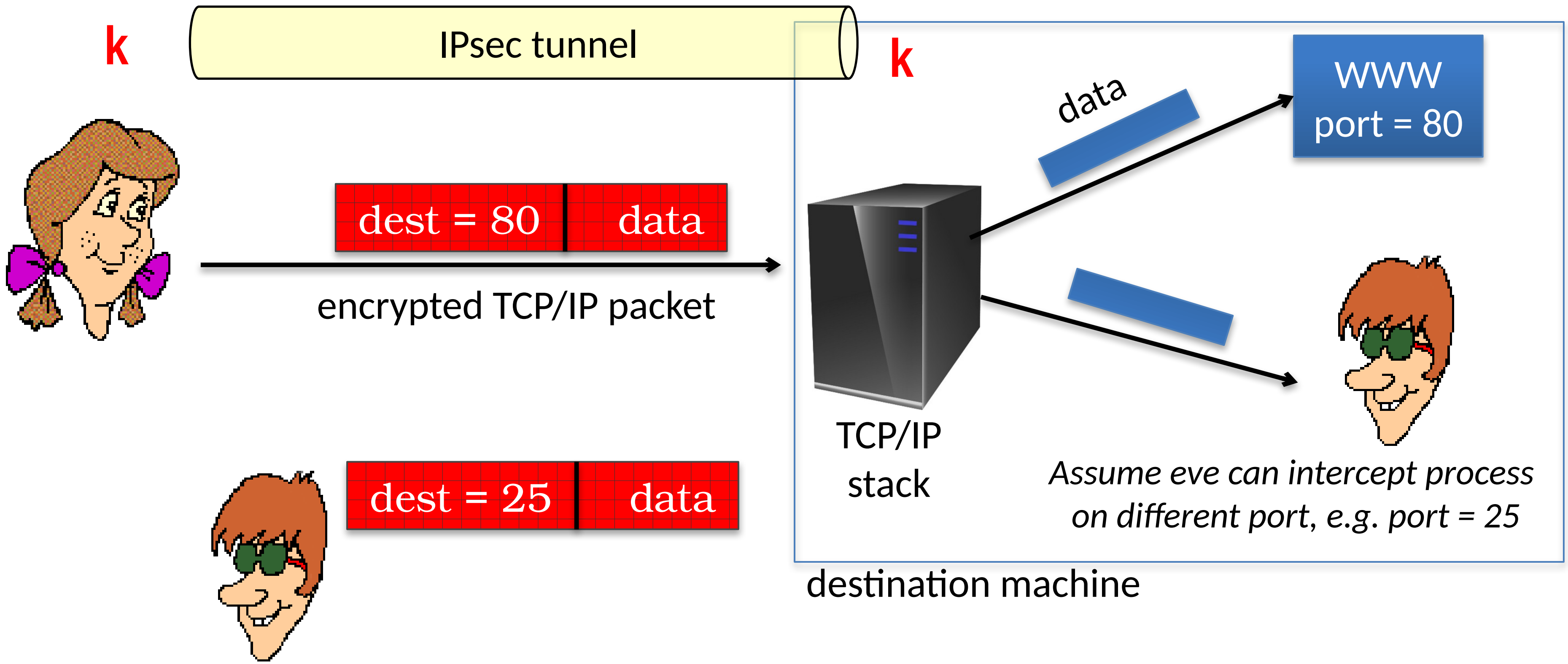
- ➔Two illustrative examples in the next slides
- ➔(credits to Dan Boneh's class)

Example 1



Souce: Dan Boneh (adapted by me)

Example 1



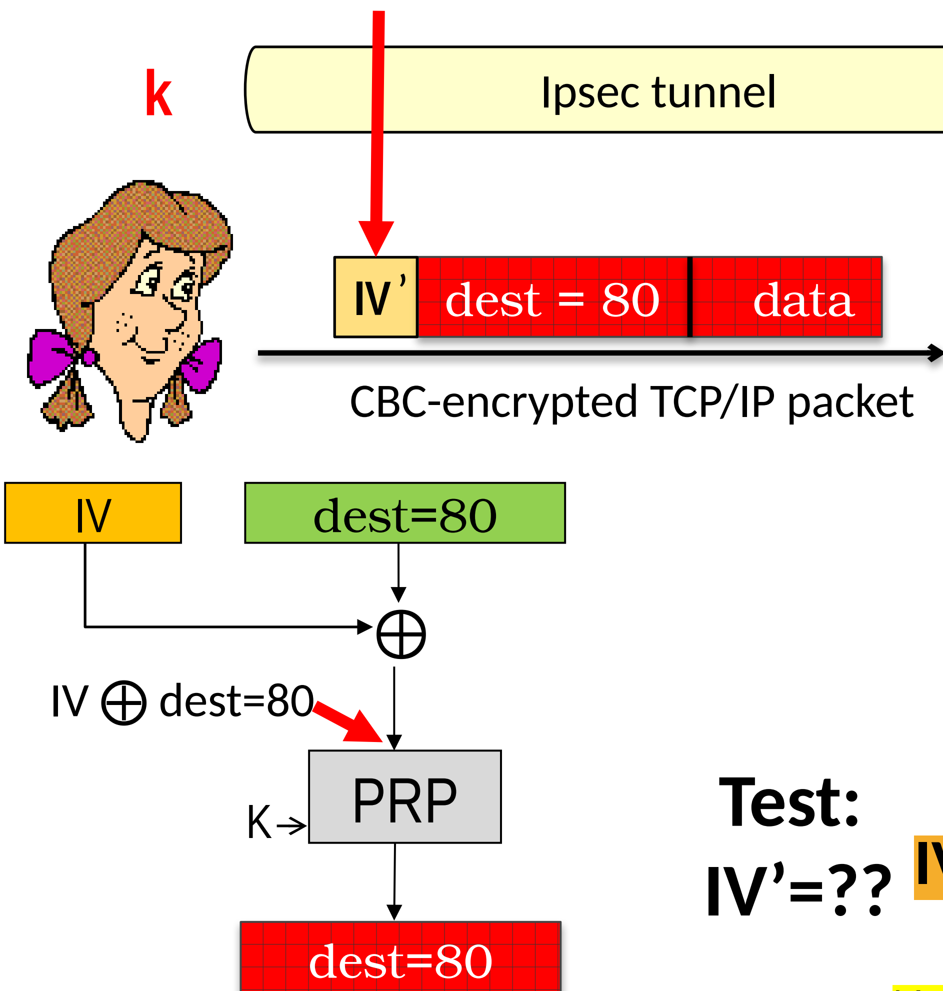
If attacker could just change port...

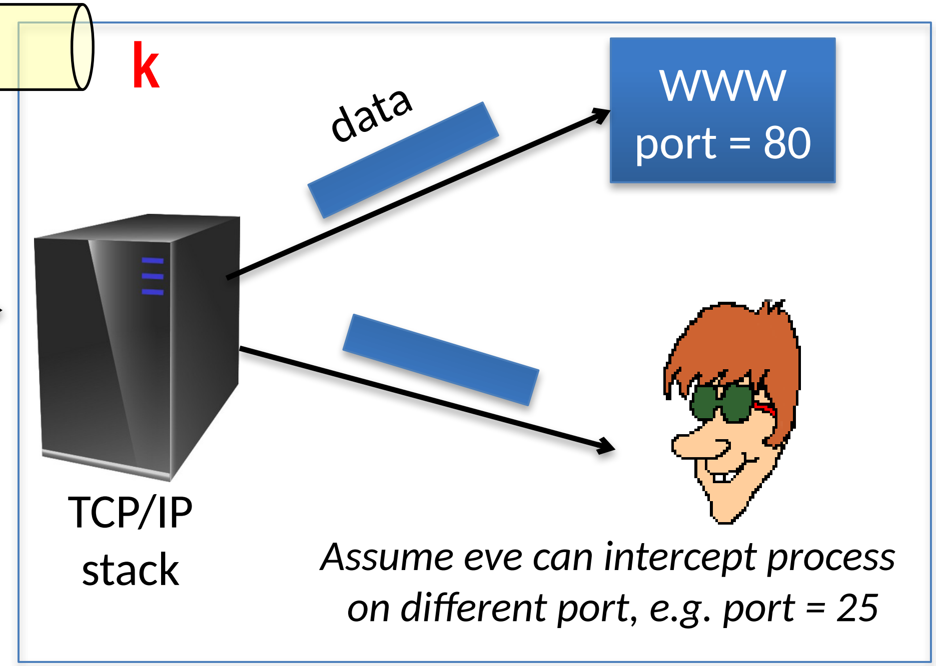
Could then read data in clear!!

TRIVIAL, with CBC encryption!!!

Souce: Dan Boneh (adapted by me)

Example 1





destination machine

Test:

IV'=?? **IV' = IV ⊕ dest=80 ⊕ dest=25**

Now the SAME previous ciphertext decrypt to a new plaintext where port=25!!

Souce: Dan Boneh (adapted by me)

I’m not yet convinced!

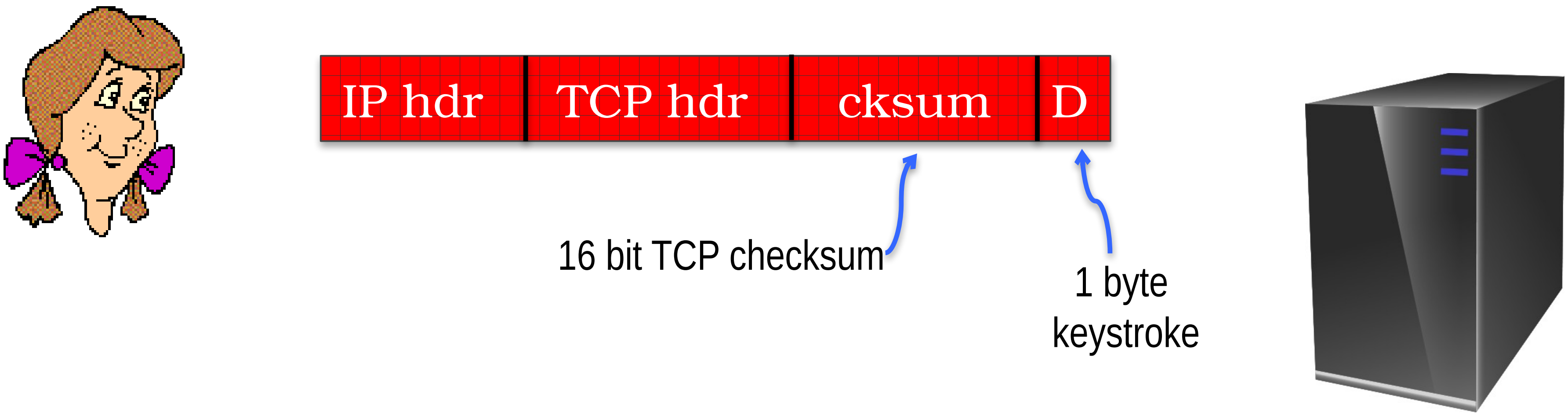
- **Too strong (?) attacker model!**
  - Attacker has access to the backend

- **And you always use CBC in your examples!**
  - Perhaps CTR is not vulnerable??

See next slide an example of CCA against CTR!

Example 2:
Attacker has only network access;

Remote terminal app: each keystroke encrypted with CTR mode



FACT: TCP/IP stack processes only valid packets

→ TCP will ACK only packets with correct cksum

(let's use this as an oracle!!)

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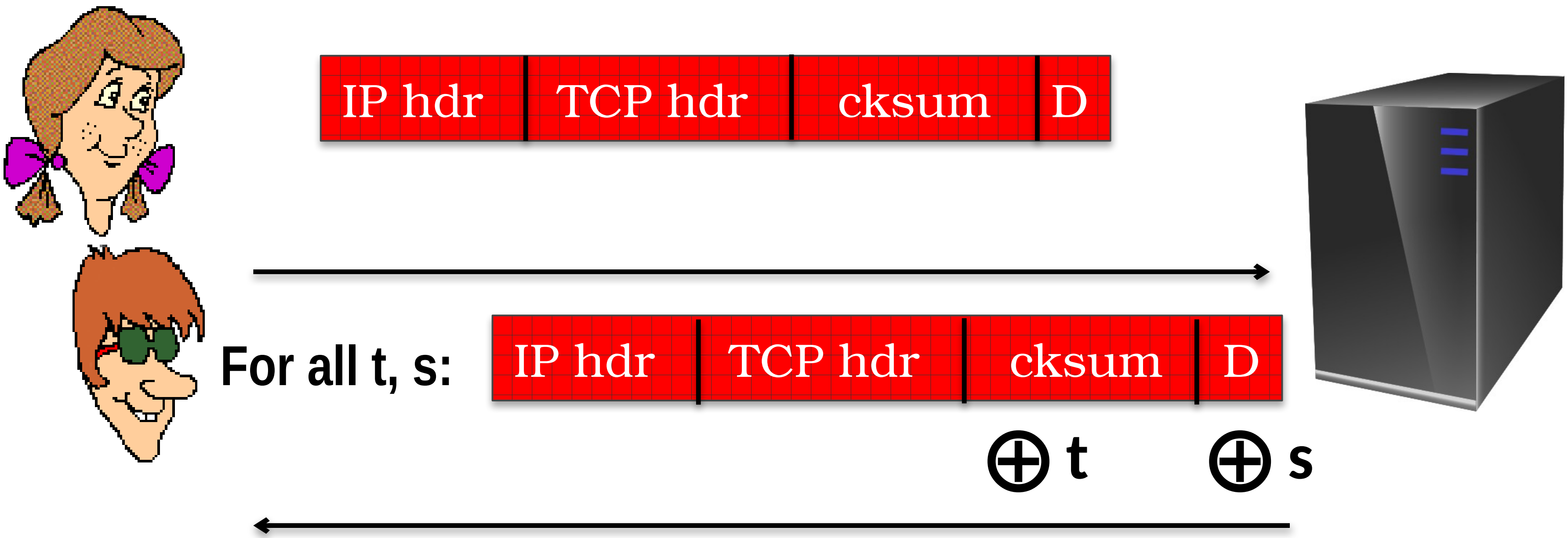


ACK if valid checksum, nothing otherwise

MANY (!) valid equations: {checksum(hdr, D⊕s) = t ⊕ checksum(hdr, D)}
hdr often known, {cksum, D} unknown → may be solved!

Souce: Dan Boneh (adapted by me)

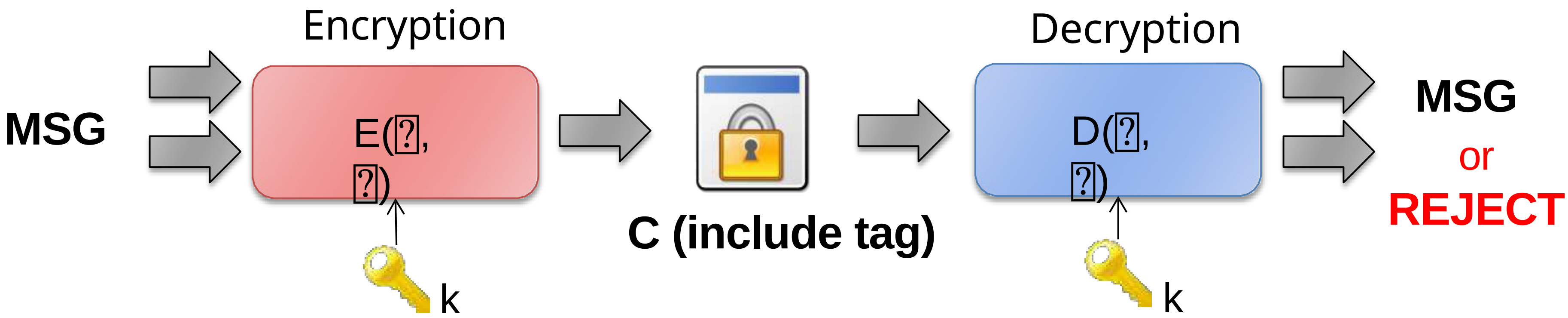
Take home

CPA security cannot guarantee secrecy under active attacks.

Only use one of two modes:

- **If message needs integrity but no confidentiality:**
  use a MAC
- **If message needs both integrity and confidentiality:**
  use authenticated encryption modes

Authenticated Encryption



Unlike standard ENC which always returns a msg, AE may output a "REJECT"

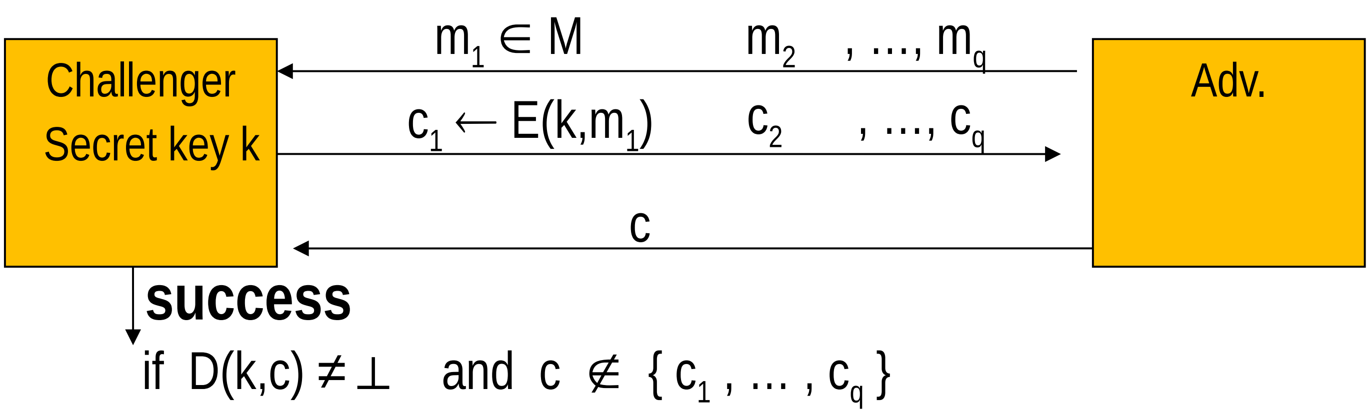
AE **fundamentally stronger** than basic cipher: decrypts ONLY IF authentication tag is valid.

→*Prevents attacker from performing CCA*

Security definition

➔AE is secure if:

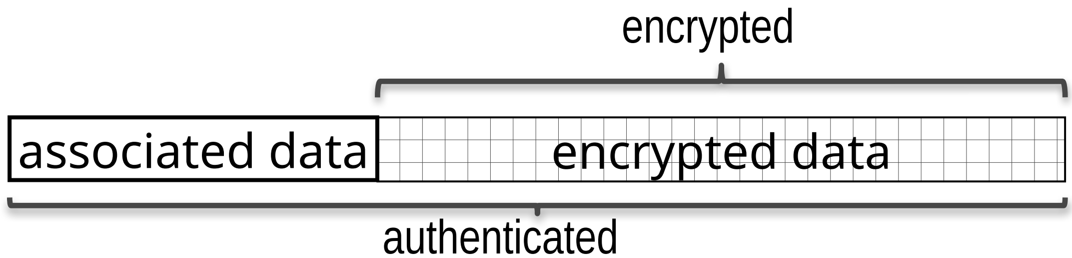
1. is semantically secure under CPA
2. Guarantees **ciphertext integrity**



<u>**ciphertext integrity**</u> (informally): if probability of success is negligible

(note: success even if forged c decrypting to completely random msg!!!)

AEAD: AE with Associated Data



- **Associated data: often packet hdr**
  - hdr must usually remain in plaintext
- **Extreme cases also frequent**
  - No associated data → CCA-secure encryption
  - No encrypted data → secure MAC
- **Now should be clear why modern protocols only use AEAD (e.g. TLSv1.3)**
  - If no AEAD, then use Encrypt-then-MAC, also CCA-secure

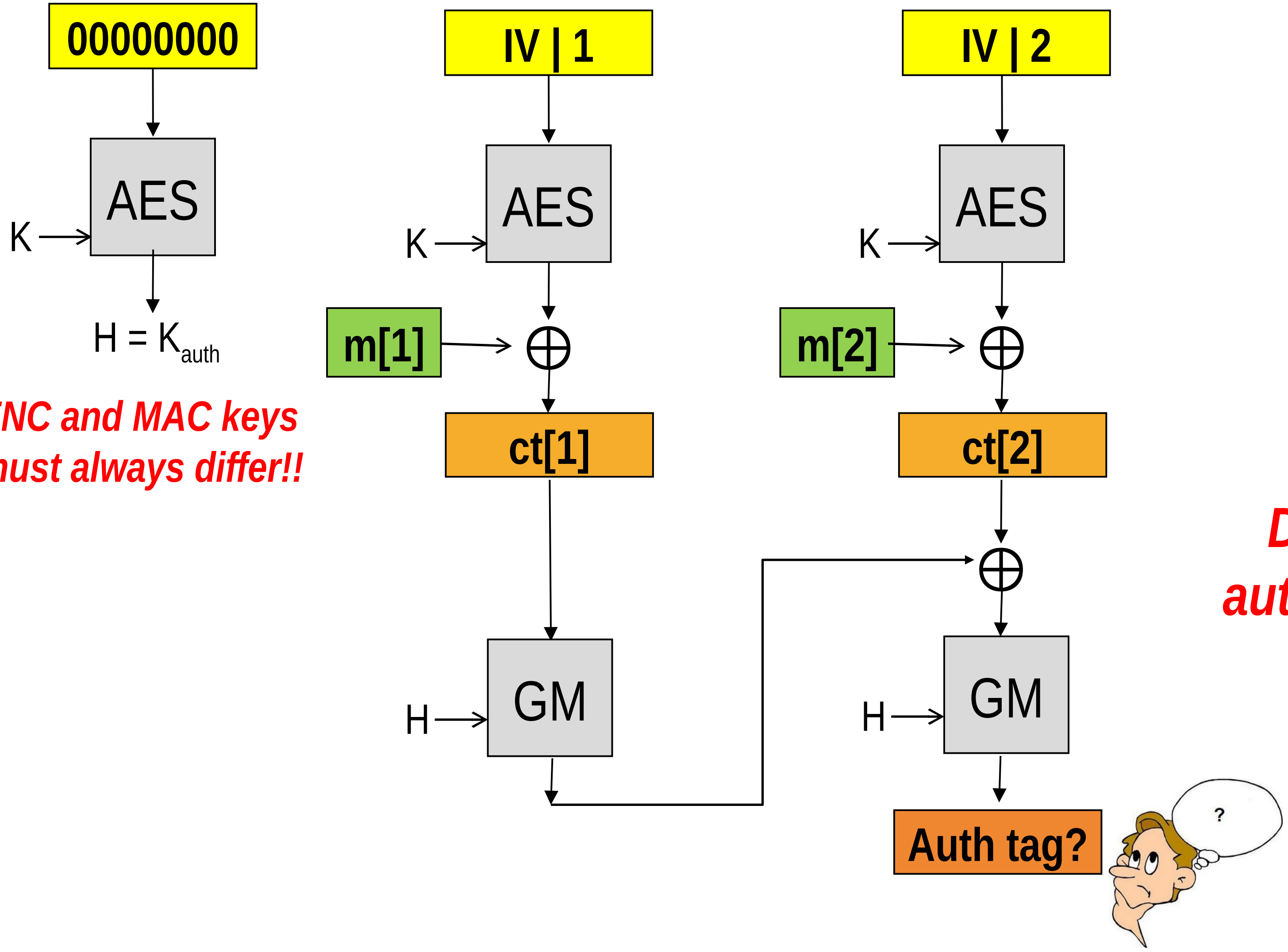
Design choices for AEAD

- **Structure**
  - Two-layer = first encrypt and then MAC (e.g. AES-GCM)
  - One-layer = all in one (e.g. OCB → faster than GCM)
- **Performance**
  - First encrypt then MAC may require two pass → slow
  - «full» parallelizability: hard
    - → authenticity check requires «all» data!
  - ***Streamability (= online cipher): one pass***
- **Functional requirements**
  - Where to place Associated Data? Before? After? Any mix?
- **Misuse resistance**
  - How robust is AEAD **when IV is reused**?

AES-GCM: Galois Counter Mode

- **First standardized algorithm**
  - NIST SP 800 38D recommendation
  - Used in most security protocols (IPsec, TLS, …)
    - (though not used in WiFi → AES-CCM)
- **Encrypt-then-MAC structure**
  - Encrypt: AES-CTR (this is the CM – Counter Mode - in GCM)
  - MAC: GHASH (this is the G – Galois - in GCM)
    - Not a crypto hash, but a (much faster!) multiplication in GF(2^{128}) with an auth key
      - more later

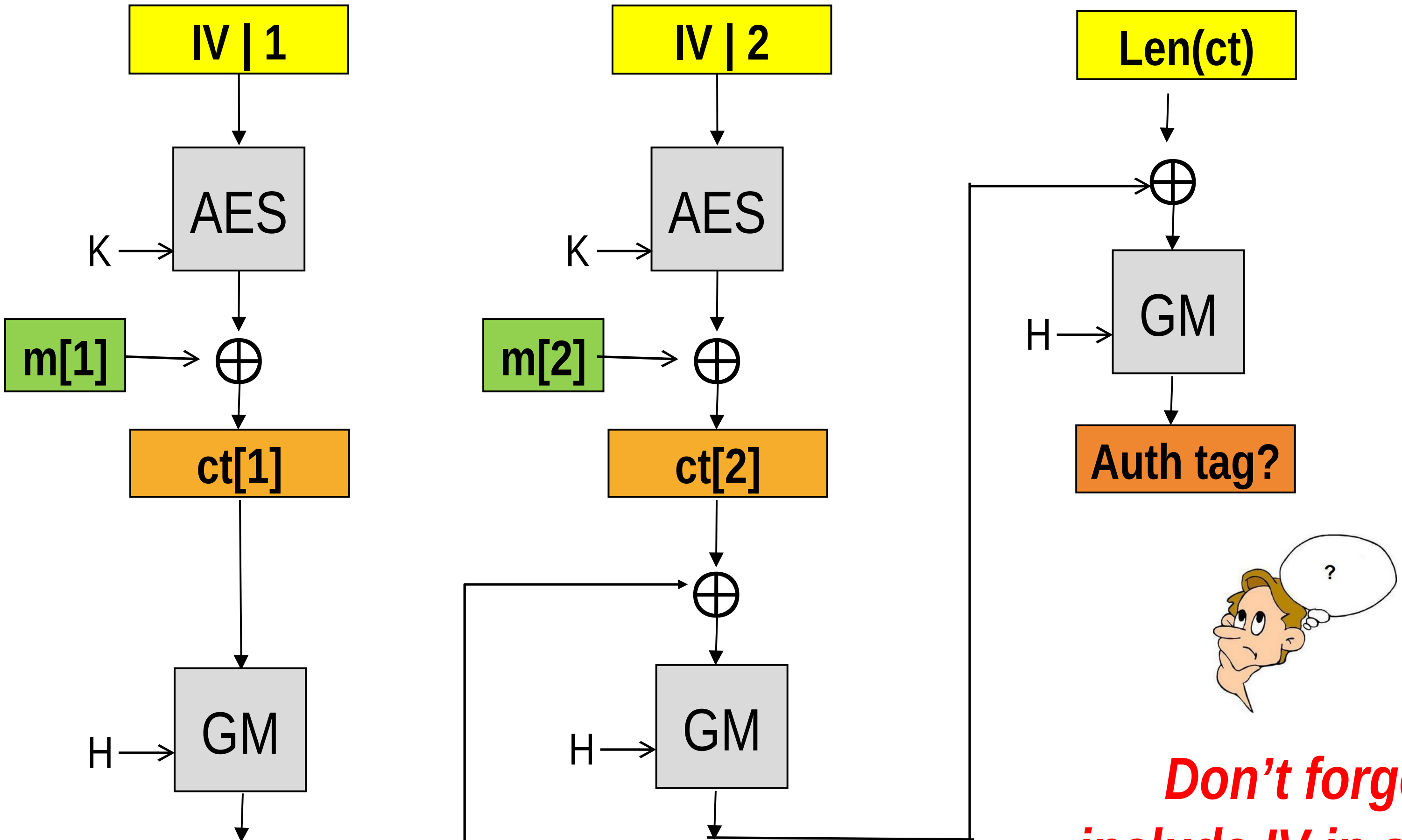
Construction @ high level (example: msg in two blocks)



ENC and MAC keys must always differ!!

Don't forget to authenticate LEN!!

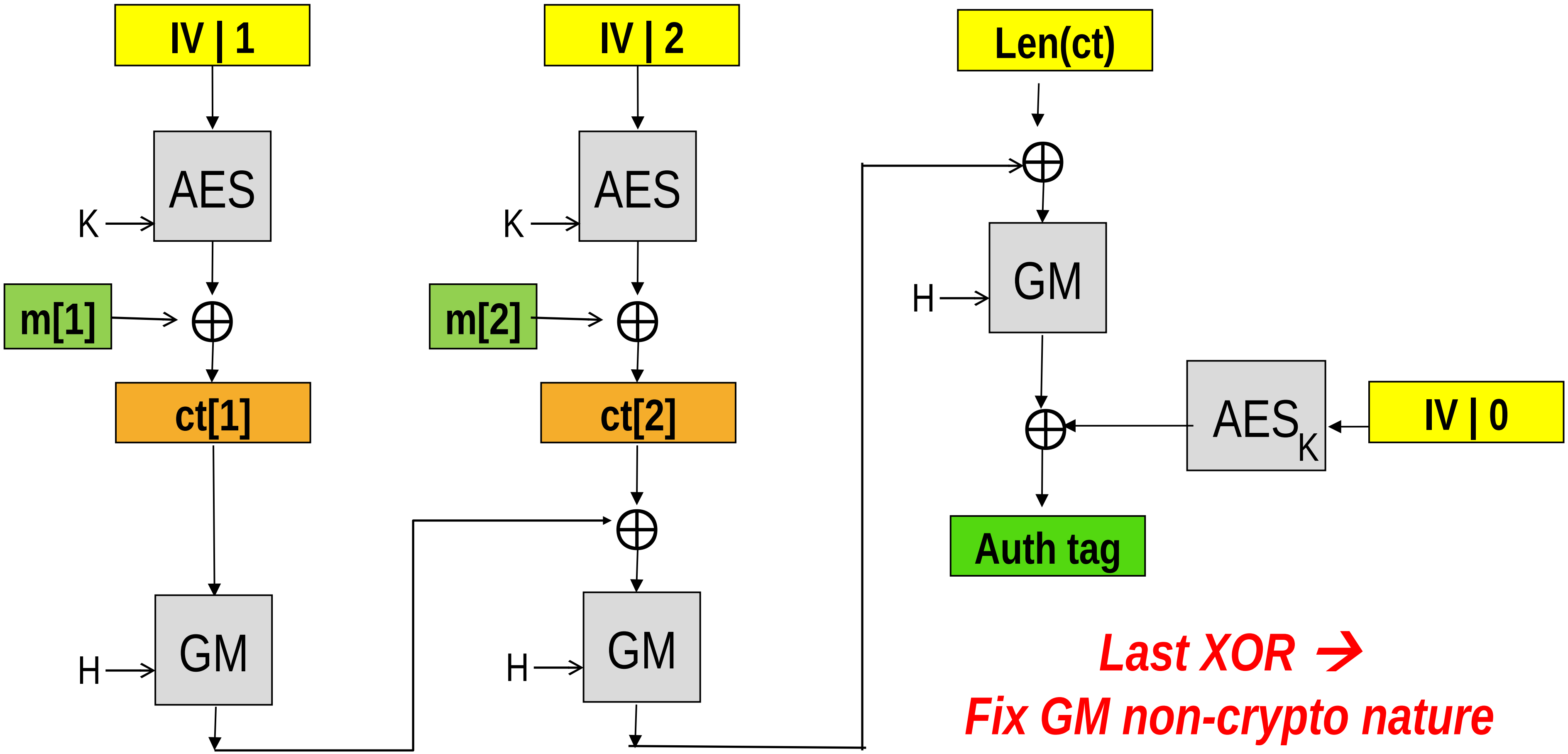
Construction @ high level (example: msg in two blocks)



Don't forget to include IV in auth tag!!

(remember previous CCA example against IV)

Construction @ high level (example: msg in two blocks)



Last XOR → Fix GM non-crypto nature (Wegman-Carter MAC) (see next slides)

Wegman-Carter MAC

- **Crypto hash are slow**
- **Non crypto hash can be a lot faster**

- **Wegman-Carter: how to MAC using NON-CRYPTO hash functions**

- **Two building blocks:**
  - Universal Hash Function
  - Pseudo Random Function

Universal Hash Function

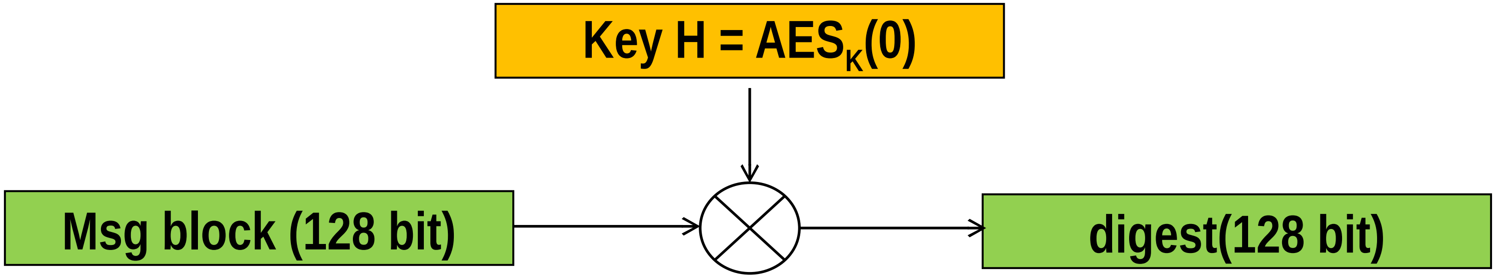
- **A Keyed Hash Function is a FAMILY of functions**
  - $H_k(msg) \rightarrow \{0,1,\ldots, m-1\}$ → k = key

- **The family is «universal» if it is hard for an attacker <u>which does NOT know the key</u> to find a collision $H_k(M_1)=H_k(M_2)$**
  - MANY differences with respect to crypto hash:
    - If you know the key, finding a collision CAN be easy
    - The output $H_k(M)$ may NOT be pseudo-random

- **Companion property:**
  - for any pair x≠y, Prob $\{ H_x(M) = H_y(M) \} \leq 1/m$
    - Changing key should randomly change digest (the probability that a collision repeats after changing key should be negligible)
    - there should be no pair (M1, M2) that gives the same hash for many different keys.

AES-GCM: GHASH

- **GHASH is NOT crypto, but it is «just» a universal hash function**



- **Building block: Multiplication in GF(2^{128})**
  - a.k.a. polynomial carry-less multiplication
    - CLMUL() instruction available in CPUs

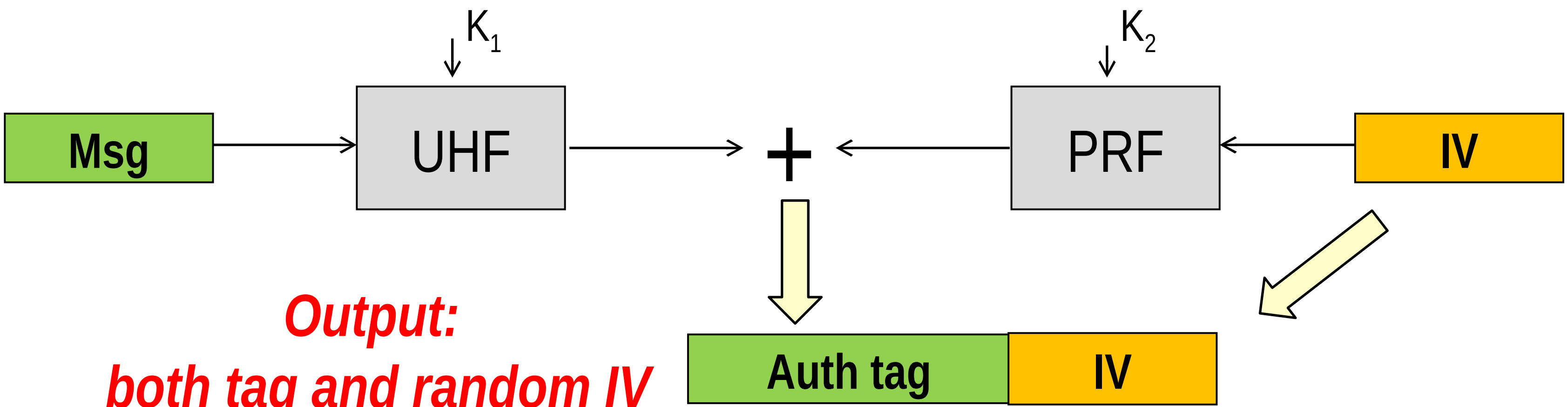
Note: multiplication in GF(2^n)

- **Polynomial multiplication**
  - Example GF(2^4):
    - **1110 x 1011** =
    - = (x^3 + x^2 + x) * (x^3 + x + 1) =
    - = $x^6 + x^5 + 2x^4 + 2x^3 + 2x^2 + x$ = (binary coeff: 1+1→0!)
    - = $x^6 + x^5 + x$ = **01100010**
  - Must now «reduce» to 4 bits: rest of division by «reduction» poly
    - Conceptually analogous to multiplication modulo prime # in standard arithmetics
  - Example GF(2^4) using irreducible poly x^4+x+1 :
    - $(x^6 + x^5 + x) / (x^4 + x + 1)$ =
    - = $x^2 + x + (-x^3 - 2x^2)/(x^4 + x + 1) = x^2 + x + (x^3)/(x^4 + x + 1)$
    - Result: rest of division: x^3 = **1000**

GHASH GF(2^{128}) irreducible poly used: $x^{128}+x^7+x^2+x+1$

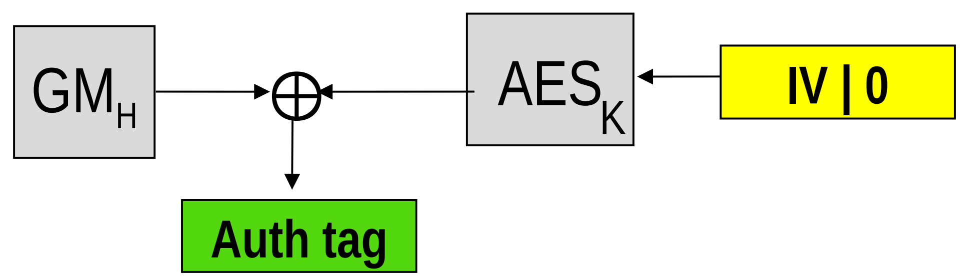
Wegman-Carter construction

➔ **A secure MAC can be constructed from an UHF and a PRF as $UHF_{k1}(msg) + PRF_{k2}(rand)$**

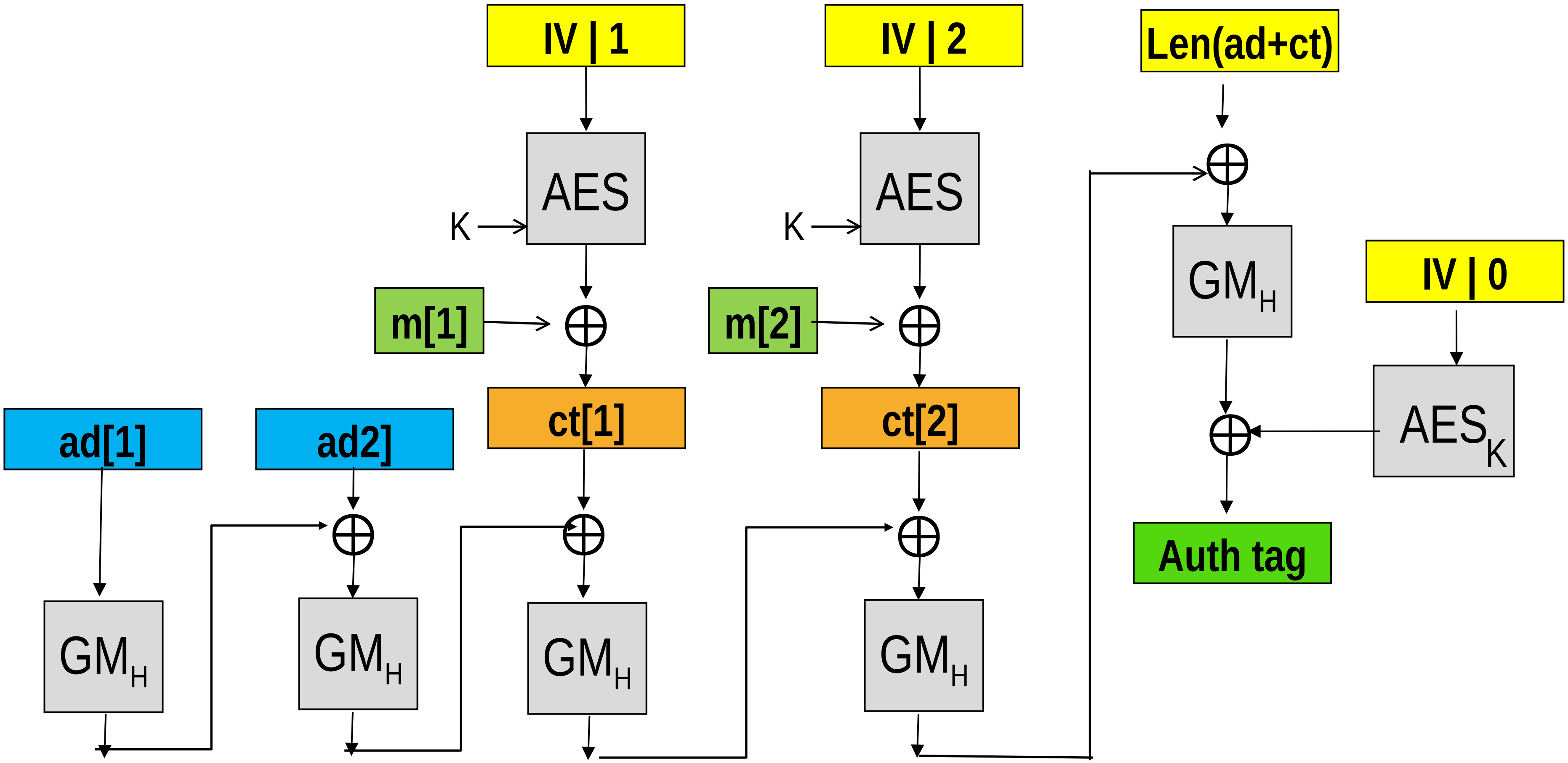


Output:
both tag and random IV
a new tool for you:
randomized MAC!

AES-GCM specific case:



What about Associated Data?



Len adjusted to include also associated data

GCM critical issue: nonce reuse

- **AES-GCM tag:**
  - tag = $GHASH_H(CT) \oplus AES_K(IV,0)$
- **Nonce reused (same IV):**
  - tag1 = $GHASH_H(CT1) \oplus AES_K(IV,0)$
  - tag2 = $GHASH_H(CT2) \oplus AES_K(IV,0)$
  - tag1 $\oplus$ tag2 = $GHASH_H(CT1) \oplus GHASH_H(CT2)$
- **But GHASH is… linear (poly mult)!**
  - Attacker can now recover auth key H, and forge valid messages!
    - (Though at least encryption key K remains unknown…)
- **There exists some mode robust to nonce reuse?**
  - Yes, AES-GCM-SIV (Synthetic IV), see RFC 8452, April 2019